```
%% GRAVITATION (MTW)
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%% Box 14.5 CURVATURE COMPUTED USING EXTERIOR DIFFERENTIAL
%% FORMS (METRIC FOR FRIEDMANN COSMOLOGY), Page 355

load_package excalc$

pform a=0$
fdomain a=a(t);

write "Define the Friedmann coframe"$
coframe o(t)      = 1                          * d t,
        o(chi)    = a                          * d chi,
        o(theta)  = a * sin(chi)               * d theta,
        o(phi)    = a * sin(chi) * sin(theta) * d phi
  with metric g = - o(t) * o(t) + o(chi) * o(chi)
                  + o(theta) * o(theta)
                  + o(phi) * o(phi)$
frame e$
displayframe;
DETM!*;

*** .*. redefined

*** × redefined

*** ^ redefined
```

Define the Friedmann coframe

$$o^t = d\,t$$

$$o^\chi = d\,\chi\,a$$

$$o^\theta = d\,\theta\,\sin(\chi)\,a$$

$$o^\phi = d\,\phi\,\sin(\chi)\,\sin(\theta)\,a$$

$$-1$$

```
on fancy$
on nero$
factor o,^$

clear omega$
riemannconx omega$
write "Display the connection form"$
omega(k,-1);
```

Display the connection form

$$\omega^{\chi}{}_{t} := \frac{o^{\chi}\,\frac{\partial\, a}{\partial\, t}}{a}$$

$$\omega^{\theta}{}_{t} := \frac{o^{\theta}\,\frac{\partial\, a}{\partial\, t}}{a}$$

$$\omega^{\phi}{}_{t} := \frac{o^{\phi}\,\frac{\partial\, a}{\partial\, t}}{a}$$

$$\omega^{t}{}_{\chi} := \frac{o^{\chi}\,\frac{\partial\, a}{\partial\, t}}{a}$$

$$\omega^{\theta}{}_{\chi} := \frac{o^{\theta}\,\cos(\chi)}{\sin(\chi)\,a}$$

$$\omega^{\phi}{}_{\chi} := \frac{o^{\phi}\,\cos(\chi)}{\sin(\chi)\,a}$$

$$\omega^{t}{}_{\theta} := \frac{o^{\theta}\,\frac{\partial\, a}{\partial\, t}}{a}$$

$$\omega^{\chi}{}_{\theta} := \frac{-o^{\theta}\,\cos(\chi)}{\sin(\chi)\,a}$$

$$\omega^{\phi}{}_{\theta} := \frac{o^{\phi}\,\cos(\theta)}{\sin(\chi)\,\sin(\theta)\,a}$$

$$\omega^{t}{}_{\phi} := \frac{o^{\phi}\,\frac{\partial\, a}{\partial\, t}}{a}$$

$$\omega^{\chi}{}_{\phi} := \frac{-o^{\phi}\,\cos(\chi)}{\sin(\chi)\,a}$$

$$\omega^{\theta}{}_{\phi} := \frac{-o^{\phi}\,\cos(\theta)}{\sin(\chi)\,\sin(\theta)\,a}$$

```
write "Display the connection form in Matrix"$
linelength 100$
coords := {t, chi, theta, phi}$
matrix Momega(4, 4)$
for k := 1:4 do for l := 1:4 do
Momega(k, l) := omega(part(coords, k), part(coords, l))$
Momega;
```

Display the connection form in Matrix

$$\begin{pmatrix} 0 & \frac{o^{\chi}\,\frac{\partial\,a}{\partial\,t}}{a} & \frac{o^{\theta}\,\frac{\partial\,a}{\partial\,t}}{a} & \frac{o^{\phi}\,\frac{\partial\,a}{\partial\,t}}{a} \\ \frac{-o^{\chi}\,\frac{\partial\,a}{\partial\,t}}{a} & 0 & \frac{-o^{\theta}\,\cos(\chi)}{\sin(\chi)\,a} & \frac{-o^{\phi}\,\cos(\chi)}{\sin(\chi)\,a} \\ \frac{-o^{\theta}\,\frac{\partial\,a}{\partial\,t}}{a} & \frac{o^{\theta}\,\cos(\chi)}{\sin(\chi)\,a} & 0 & \frac{-o^{\phi}\,\cos(\theta)}{\sin(\chi)\,\sin(\theta)\,a} \\ \frac{-o^{\phi}\,\frac{\partial\,a}{\partial\,t}}{a} & \frac{o^{\phi}\,\cos(\chi)}{\sin(\chi)\,a} & \frac{o^{\phi}\,\cos(\theta)}{\sin(\chi)\,\sin(\theta)\,a} & 0 \end{pmatrix}$$

```
clear curv,riemann,ricci,riccisc$
pform curv(k,l)=2,{riemann(a,b,c,d),ricci(a,b),riccisc}=0$

index_symmetries curv(k,l): antisymmetric,
                 riemann(k,l,m,n): antisymmetric in {k,l},{m,n}
                                   symmetric in {{k,l},{m,n}},
                 ricci(k,l): symmetric;

write "Display the curvature form"$
curv(k,-l) := d omega(k,-l) + omega(k,-m) ^ omega(m,-l);
```

Display the curvature form

$$\mathrm{curv}^{\chi}{}_{t} := \frac{o^{t} \wedge o^{\chi}\,\frac{\partial^{2}\,a}{\partial\,t^{2}}}{a}$$

$$\mathrm{curv}^{\theta}{}_{t} := \frac{o^{t} \wedge o^{\theta}\,\frac{\partial^{2}\,a}{\partial\,t^{2}}}{a}$$

$$\mathrm{curv}^{\phi}{}_{t} := \frac{o^{t} \wedge o^{\phi}\,\frac{\partial^{2}\,a}{\partial\,t^{2}}}{a}$$

$$\mathrm{curv}^{t}{}_{\chi} := \frac{o^{t} \wedge o^{\chi}\,\frac{\partial^{2}\,a}{\partial\,t^{2}}}{a}$$

$$\mathrm{curv}^{\theta}{}_{\chi} := \frac{-o^{\chi} \wedge o^{\theta}\,\left(\left(\frac{\partial\,a}{\partial\,t}\right)^{2} + 1\right)}{a^{2}}$$

$$\mathrm{curv}^{\phi}{}_{\chi} := \frac{-o^{\chi} \wedge o^{\phi}\,\left(\left(\frac{\partial\,a}{\partial\,t}\right)^{2} + 1\right)}{a^{2}}$$

$$\mathrm{curv}^{t}{}_{\theta} := \frac{o^{t} \wedge o^{\theta}\,\frac{\partial^{2}\,a}{\partial\,t^{2}}}{a}$$

$$\mathrm{curv}^{\chi}{}_{\theta} := \frac{o^{\chi} \wedge o^{\theta}\,\left(\left(\frac{\partial\,a}{\partial\,t}\right)^{2} + 1\right)}{a^{2}}$$

$$\mathrm{curv}^{\phi}{}_{\theta} := \frac{o^{\theta} \wedge o^{\phi}\,\left(\cos(\chi)^{2} - \left(\frac{\partial\,a}{\partial\,t}\right)^{2}\,\sin(\chi)^{2} - 1\right)}{\sin(\chi)^{2}\,a^{2}}$$

$$\mathrm{curv}^{t}{}_{\phi} := \frac{o^{t} \wedge o^{\phi}\, \frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{curv}^{\chi}{}_{\phi} := \frac{o^{\chi} \wedge o^{\phi}\, \left(\left(\frac{\partial a}{\partial t}\right)^2 + 1\right)}{a^2}$$

$$\mathrm{curv}^{\theta}{}_{\phi} := \frac{o^{\theta} \wedge o^{\phi}\, \left(-\cos(\chi)^2 + \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 + 1\right)}{\sin(\chi)^2\, a^2}$$

```
write "Display the curvature form in Matrix"$
linelength 200$
matrix Mcurv(4, 4)$
for k := 1:4 do for l := 1:4 do
Mcurv(k, l) := curv(part(coords, k), part(coords, l))$
Mcurv;
```

Display the curvature form in Matrix

$$\begin{pmatrix} 0 & \frac{o^{t}\wedge o^{\chi}\, \frac{\partial^2 a}{\partial t^2}}{a} & \frac{o^{t}\wedge o^{\theta}\, \frac{\partial^2 a}{\partial t^2}}{a} & \frac{o^{t}\wedge o^{\phi}\, \frac{\partial^2 a}{\partial t^2}}{a} \\ \frac{-o^{t}\wedge o^{\chi}\, \frac{\partial^2 a}{\partial t^2}}{a} & 0 & \frac{o^{\chi}\wedge o^{\theta}\, \left(\left(\frac{\partial a}{\partial t}\right)^2+1\right)}{a^2} & \frac{o^{\chi}\wedge o^{\phi}\, \left(\left(\frac{\partial a}{\partial t}\right)^2+1\right)}{a^2} \\ \frac{-o^{t}\wedge o^{\theta}\, \frac{\partial^2 a}{\partial t^2}}{a} & \frac{-o^{\chi}\wedge o^{\theta}\, \left(\left(\frac{\partial a}{\partial t}\right)^2+1\right)}{a^2} & 0 & \frac{o^{\theta}\wedge o^{\phi}\, \left(-\cos(\chi)^2+\left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2+1\right)}{\sin(\chi)^2\, a^2} \\ \frac{-o^{t}\wedge o^{\phi}\, \frac{\partial^2 a}{\partial t^2}}{a} & \frac{-o^{\chi}\wedge o^{\phi}\, \left(\left(\frac{\partial a}{\partial t}\right)^2+1\right)}{a^2} & \frac{o^{\theta}\wedge o^{\phi}\, \left(\cos(\chi)^2-\left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2-1\right)}{\sin(\chi)^2\, a^2} & 0 \end{pmatrix}$$

```
write "Display the Riemann Tensor"$
riemann(a,b,c,d) := e(d) _| ( e(c) _| curv(a,b) );
riemann(a,-b,-c,-d);
riemann(-a,-b,-c,-d);
```

Display the Riemann Tensor

$$\mathrm{riemann}^{t\,\chi\,t\,\chi} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t\,\theta\,t\,\theta} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\chi\,\theta\,\chi\,\theta} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}^{t\,\phi\,t\,\phi} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\chi\,\phi\,\chi\,\phi} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}^{\theta\,\phi\,\theta\,\phi} := \frac{-\cos(\chi)^2 + \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 + 1}{\sin(\chi)^2\, a^2}$$

$$\mathrm{riemann}^{\chi}{}_{t\,\chi\,t} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t}{}_{\chi\,\chi\,t} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\theta}{}_{t\,\theta\,t} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t}{}_{\theta\,\theta\,t} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\phi}{}_{t\,\phi\,t} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t}{}_{\phi\,\phi\,t} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\chi}{}_{t\,t\,\chi} := \frac{\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t}{}_{\chi\,t\,\chi} := \frac{\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\theta}{}_{\chi\,\theta\,\chi} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}^{\chi}{}_{\theta\,\theta\,\chi} := \frac{-\left(\left(\frac{\partial a}{\partial t}\right)^2 + 1\right)}{a^2}$$

$$\mathrm{riemann}^{\phi}{}_{\chi\,\phi\,\chi} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}^{\chi}{}_{\phi\,\phi\,\chi} := \frac{-\left(\left(\frac{\partial a}{\partial t}\right)^2 + 1\right)}{a^2}$$

$$\mathrm{riemann}^{\theta}{}_{t\,t\,\theta} := \frac{\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t}{}_{\theta\,t\,\theta} := \frac{\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\theta}{}_{\chi\,\chi\,\theta} := \frac{-\left(\left(\frac{\partial a}{\partial t}\right)^2 + 1\right)}{a^2}$$

$$\mathrm{riemann}^{\chi}{}_{\theta\,\chi\,\theta} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}^{\phi}{}_{\theta\,\phi\,\theta} := \frac{-\cos(\chi)^2 + \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 + 1}{\sin(\chi)^2 a^2}$$

$$\mathrm{riemann}^{\theta}{}_{\phi\,\phi\,\theta} := \frac{\cos(\chi)^2 - \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 - 1}{\sin(\chi)^2 a^2}$$

$$\mathrm{riemann}^{\phi}{}_{t\,t\,\phi} := \frac{\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{t}{}_{\phi\,t\,\phi} := \frac{\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}^{\phi}{}_{\chi\,\chi\,\phi} := \frac{-\left(\left(\frac{\partial a}{\partial t}\right)^2 + 1\right)}{a^2}$$

$$\mathrm{riemann}^{\chi}{}_{\phi\,\chi\,\phi} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}^{\phi}{}_{\theta\,\theta\,\phi} := \frac{\cos(\chi)^2 - \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 - 1}{\sin(\chi)^2 a^2}$$

$$\mathrm{riemann}^{\theta}{}_{\phi\,\theta\,\phi} := \frac{-\cos(\chi)^2 + \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 + 1}{\sin(\chi)^2 a^2}$$

$$\mathrm{riemann}_{t\,\chi\,t\,\chi} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}_{t\,\theta\,t\,\theta} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}_{\chi\,\theta\,\chi\,\theta} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}_{t\,\phi\,t\,\phi} := \frac{-\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{riemann}_{\chi\,\phi\,\chi\,\phi} := \frac{\left(\frac{\partial a}{\partial t}\right)^2 + 1}{a^2}$$

$$\mathrm{riemann}_{\theta\,\phi\,\theta\,\phi} := \frac{-\cos(\chi)^2 + \left(\frac{\partial a}{\partial t}\right)^2 \sin(\chi)^2 + 1}{\sin(\chi)^2 a^2}$$

```
write "Display the Ricci Tensor"$
ricci(-a,-b) := riemann(c,-a,-d,-b) * g(-c,d);
```

Display the Ricci Tensor

$$\mathrm{ricci}_{t\,t} := \frac{-3\,\frac{\partial^2 a}{\partial t^2}}{a}$$

$$\mathrm{ricci}_{\chi\,\chi} := \frac{\frac{\partial^2 a}{\partial t^2}\,a + 2\left(\frac{\partial a}{\partial t}\right)^2 + 2}{a^2}$$

$$\mathrm{ricci}_{\theta\,\theta} := \frac{-\cos(\chi)^2 + \frac{\partial^2 a}{\partial t^2}\sin(\chi)^2\,a + 2\left(\frac{\partial a}{\partial t}\right)^2\sin(\chi)^2 + \sin(\chi)^2 + 1}{\sin(\chi)^2\,a^2}$$

$$\mathrm{ricci}_{\phi\,\phi} := \frac{-\cos(\chi)^2 + \frac{\partial^2 a}{\partial t^2}\sin(\chi)^2\,a + 2\left(\frac{\partial a}{\partial t}\right)^2\sin(\chi)^2 + \sin(\chi)^2 + 1}{\sin(\chi)^2\,a^2}$$

```
write "Display the Ricci Scalar"$
riccisc := ricci(-a,-b) * g(a,b);
```

Display the Ricci Scalar

$$\mathrm{riccisc} := \frac{2\left(-\cos(\chi)^2 + 3\,\frac{\partial^2 a}{\partial t^2}\sin(\chi)^2\,a + 3\left(\frac{\partial a}{\partial t}\right)^2\sin(\chi)^2 + 2\,\sin(\chi)^2 + 1\right)}{\sin(\chi)^2\,a^2}$$

```
write "Display the Einstein Tensor"$
clear einstein$
pform einstein(a)=3$
einstein(-a) := (1/2) * curv(b,c) ^ #( o(-b) ^ o(-c) ^ o(-a) );
```

Display the Einstein Tensor

$$\mathrm{einstein}_t := \frac{o^\chi \wedge o^\theta \wedge o^\phi\left(-\cos(\chi)^2 + 3\left(\frac{\partial a}{\partial t}\right)^2\sin(\chi)^2 + 2\,\sin(\chi)^2 + 1\right)}{\sin(\chi)^2\,a^2}$$

$$\mathrm{einstein}_\chi := \frac{o^t \wedge o^\theta \wedge o^\phi\left(\cos(\chi)^2 - 2\,\frac{\partial^2 a}{\partial t^2}\sin(\chi)^2\,a - \left(\frac{\partial a}{\partial t}\right)^2\sin(\chi)^2 - 1\right)}{\sin(\chi)^2\,a^2}$$

$$\mathrm{einstein}_\theta := \frac{o^t \wedge o^\chi \wedge o^\phi\left(2\,\frac{\partial^2 a}{\partial t^2}\,a + \left(\frac{\partial a}{\partial t}\right)^2 + 1\right)}{a^2}$$

$$\mathrm{einstein}_\phi := \frac{o^t \wedge o^\chi \wedge o^\theta\left(-2\,\frac{\partial^2 a}{\partial t^2}\,a - \left(\frac{\partial a}{\partial t}\right)^2 - 1\right)}{a^2}$$

```
% Define the velocity and acceleration components.
% Those should bear contravariant indices.
pform {u(k),a(k)} = 0;
remflag('(u a),'covariant);
factor u,a;

write "Display the Geodesic equations"$
```

```
a(k) := e(-m) _| omega(-l, k) * u(l) * u(m);
a(-k);
```

Display the Geodesic equations

$$a^t := \frac{-(u^\chi)^2 \frac{\partial a}{\partial t} - (u^\phi)^2 \frac{\partial a}{\partial t} - (u^\theta)^2 \frac{\partial a}{\partial t}}{a}$$

$$a^\chi := \frac{-u^\chi u^t \frac{\partial a}{\partial t} \sin(\chi) + (u^\phi)^2 \cos(\chi) + (u^\theta)^2 \cos(\chi)}{a \sin(\chi)}$$

$$a^\theta := \frac{-u^\chi u^\theta \cos(\chi) \sin(\theta) + (u^\phi)^2 \cos(\theta) - u^t u^\theta \frac{\partial a}{\partial t} \sin(\chi) \sin(\theta)}{a \sin(\chi) \sin(\theta)}$$

$$a^\phi := \frac{-u^\chi u^\phi \cos(\chi) \sin(\theta) - u^\phi u^t \frac{\partial a}{\partial t} \sin(\chi) \sin(\theta) - u^\phi u^\theta \cos(\theta)}{a \sin(\chi) \sin(\theta)}$$

$$a_t := \frac{(u^\chi)^2 \frac{\partial a}{\partial t} + (u^\phi)^2 \frac{\partial a}{\partial t} + (u^\theta)^2 \frac{\partial a}{\partial t}}{a}$$

$$a_\chi := \frac{-u^\chi u^t \frac{\partial a}{\partial t} \sin(\chi) + (u^\phi)^2 \cos(\chi) + (u^\theta)^2 \cos(\chi)}{a \sin(\chi)}$$

$$a_\theta := \frac{-u^\chi u^\theta \cos(\chi) \sin(\theta) + (u^\phi)^2 \cos(\theta) - u^t u^\theta \frac{\partial a}{\partial t} \sin(\chi) \sin(\theta)}{a \sin(\chi) \sin(\theta)}$$

$$a_\phi := \frac{-u^\chi u^\phi \cos(\chi) \sin(\theta) - u^\phi u^t \frac{\partial a}{\partial t} \sin(\chi) \sin(\theta) - u^\phi u^\theta \cos(\theta)}{a \sin(\chi) \sin(\theta)}$$

```
b := killing_vector vec$
write "Killing Vectors Formal expression"$
part(b,1);
write "PDEs for the coefficients"$
for k := 1 step 1 until length(part(b,2)) do write part(part(b,2),k);

showtime;
end;
```

Killing Vectors Formal expression

$$e_\chi \, \mathrm{vec}^\chi + e_\phi \, \mathrm{vec}^\phi + e_t \, \mathrm{vec}^t + e_\theta \, \mathrm{vec}^\theta$$

PDEs for the coefficients

$$\frac{\partial \operatorname{vec}^t}{\partial t}$$

$$\frac{\partial \operatorname{vec}^\chi}{\partial t} a - \frac{\partial \operatorname{vec}^t}{\partial \chi} - \frac{\partial a}{\partial t} \operatorname{vec}^\chi$$

$$\frac{\partial \operatorname{vec}^t}{\partial \theta} - \frac{\partial \operatorname{vec}^\theta}{\partial t} \sin(\chi) a + \frac{\partial a}{\partial t} \sin(\chi) \operatorname{vec}^\theta$$

$$\frac{\partial \operatorname{vec}^\phi}{\partial t} \sin(\chi) \sin(\theta) a - \frac{\partial \operatorname{vec}^t}{\partial \phi} - \frac{\partial a}{\partial t} \sin(\chi) \sin(\theta) \operatorname{vec}^\phi$$

$$\frac{\partial \operatorname{vec}^\chi}{\partial \chi} + \frac{\partial a}{\partial t} \operatorname{vec}^t$$

$$\cos(\chi) \operatorname{vec}^\theta - \frac{\partial \operatorname{vec}^\chi}{\partial \theta} - \frac{\partial \operatorname{vec}^\theta}{\partial \chi} \sin(\chi)$$

$$\cos(\chi) \sin(\theta) \operatorname{vec}^\phi - \frac{\partial \operatorname{vec}^\chi}{\partial \phi} - \frac{\partial \operatorname{vec}^\phi}{\partial \chi} \sin(\chi) \sin(\theta)$$

$$\cos(\chi) \operatorname{vec}^\chi + \frac{\partial \operatorname{vec}^\theta}{\partial \theta} + \frac{\partial a}{\partial t} \sin(\chi) \operatorname{vec}^t$$

$$\cos(\theta) \operatorname{vec}^\phi - \frac{\partial \operatorname{vec}^\phi}{\partial \theta} \sin(\theta) - \frac{\partial \operatorname{vec}^\theta}{\partial \phi}$$

$$\cos(\chi) \sin(\theta) \operatorname{vec}^\chi + \cos(\theta) \operatorname{vec}^\theta + \frac{\partial \operatorname{vec}^\phi}{\partial \phi} + \frac{\partial a}{\partial t} \sin(\chi) \sin(\theta) \operatorname{vec}^t$$

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Time: 190022 ms  plus GC time: 7 ms
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